

GRADE: VIII	NAME: A SQUARE AND A CUBE
LESSON NO. : 1	
No. of Sessions/Periods	12
Lesson Overview	What is the focus of the Unit? To help students understand the concept of square numbers and cube numbers. To help students identify patterns in perfect squares and cubes. To help students understand properties of perfect squares. To help students determine whether a number is a perfect square using different methods. To help students find square roots of numbers. To help students estimate square roots of non-perfect squares. To introduce cubic numbers and understand the concept of cubes. To develop logical reasoning and pattern recognition. To apply the concept of squares and cubes in solving mathematical problems.
Previous Knowledge	What prior knowledge the students need in order to be successful with this unit? Multiplication of numbers Factors and multiples Prime factorization Basic number operations Understanding of area of square Basic knowledge of exponents
Vocabulary /Terminology	What is the new vocabulary / terminology used in this lesson? Perfect square – A number obtained by multiplying a number by itself. Square root – A number which when multiplied by itself gives the given number. Perfect cube – A number obtained by multiplying a number three times by itself. Cube root – A number which when multiplied three times gives the given number. Prime factorization – Expressing a number as a product of prime numbers.
SESSION NO. : 1	CONCEPT NAME: Introduction to Squares
Learning Outcomes	Students will understand what square numbers are. Students will identify square numbers using patterns. Students will relate square numbers to the area of squares.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	NCERT Ganita Prakash Textbook Graph paper Square tiles
Teacher Activity Time Assigned: 35 min	Introduction: Provocation/Recapitulation/Preassessment Teacher begins the class with a puzzle of 100 lockers and explains how lockers are opened and closed. Teacher explains that lockers with square numbers remain open. Teacher introduces the concept of square numbers . Examples: $1 \times 1 = 1$ $2 \times 2 = 4$ $3 \times 3 = 9$ $4 \times 4 = 16$ Teacher shows square shapes using grid paper.

	Methodology: Logical thinking Visual representation
Student Activity Time Assigned: 35 min	Students observe patterns in square numbers. Students draw squares using graph paper and count the unit squares. Students list first 10 square numbers.
Assessment	Teacher asks students to identify square numbers from a list.
Assignment/HW	Learn squares from 1 to 20
Skills/Competencies	Pattern recognition Logical reasoning
Intelligences Addressed	Visual-spatial intelligence
SESSION NO. : 2	CONCEPT NAME: Properties and Patterns of Perfect Squares
Learning Outcomes	Students will recognize patterns in square numbers. Students will identify the possible unit digits of square numbers.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook Worksheet
Teacher Activity Time Assigned: 15 min	Introduction: Provocation/Recapitulation/Preassessment Teacher asks students to calculate squares of numbers from 1 to 20. Teacher explains patterns in unit digits. Teacher explains that square numbers can end with 0,1,4,5,6,9 only. Teacher demonstrates examples. Methodology: Analytical thinking
Student Activity Time Assigned: 15 min	Students fill a table of squares from 1 to 30. Students identify patterns in last digits.
Assessment	Teacher asks students to identify numbers that cannot be perfect squares.
Assignment/HW	To practice questions based on patterns
Skills/Competencies	Analytical thinking
Intelligences Addressed	Logical intelligence
SESSION NO. : 3	CONCEPT NAME: Perfect Squares and Odd Numbers
Learning Outcomes	Students will understand that square numbers can be expressed as sum of consecutive odd numbers.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook Board illustration
Teacher Activity Time Assigned:	Introduction: Provocation/Recapitulation/Preassessment Teacher explains the pattern: $1 = 1^2$ $1 + 3 = 4$

	$1 + 3 + 5 = 9$ $1 + 3 + 5 + 7 = 16$ Teacher explains that sum of first n odd numbers = n^2 . Methodology: Logical reasoning
Student Activity Time Assigned: 15 min	Students verify the pattern using examples. Students find squares using addition of odd numbers.
Assessment	Students check whether 25 and 36 are perfect squares using the odd number method.
Assignment/HW	Check whether 100, 220, 300 are perfect squares or not
Skills/Competencies	Logical reasoning
Intelligences Addressed	Mathematical intelligence
SESSION NO. : 4	CONCEPT NAME: Square Roots
Learning Outcomes	Students will understand the meaning of square roots. Students will find square roots of perfect squares.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook
Teacher Activity Time Assigned:	Introduction: Provocation/Recapitulation/Preassessment Teacher explains square root concept: $49 = 7 \times 7$ So $\sqrt{49} = 7$ Teacher explains positive and negative roots. Teacher solves examples from textbook. Methodology: Concept explanation
Student Activity Time Assigned: 15 min	Students solve square root problems.
Assessment	Students find square roots of: 64 81 100
Assignment/HW	Find the square root of 625, 1225
Skills/Competencies	Numerical reasoning
Intelligences Addressed	Logical intelligence
SESSION NO. : 5	CONCEPT NAME: Checking Perfect Squares using Prime Factorization

Learning Outcomes	Students will determine whether a number is a perfect square using prime factorization.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook
Teacher Activity Time Assigned:	Introduction: Provocation/Recapitulation/Preassessment Teacher explains method: Example $324 = 2 \times 2 \times 3 \times 3 \times 3$ Pairing factors shows it is a perfect square. Teacher demonstrates several examples.
	Methodology: Analytical reasoning
Student Activity Time Assigned: 15 min	Students perform prime factorization of numbers.
Assessment	Check whether: 156, 1156 are perfect squares.
Assignment/HW	Check whether 1152, 2187 are perfect squares or not
Skills/Competencies	Critical thinking
Intelligences Addressed	Logical intelligence
SESSION NO. : 6	CONCEPT NAME: Estimating Square Root
Learning Outcomes	Students will estimate square roots of non-perfect squares.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook
Teacher Activity Time Assigned:	Introduction: Provocation/Recapitulation/Preassessment Teacher explains estimation using nearest squares. Example: 250 lies between 225 (15^2) and 256 (16^2) Therefore $\sqrt{250} \approx 15.8$
	Methodology: Reasoning and estimation
Student Activity Time Assigned: 15 min	Students estimate square roots of numbers like 125, 200.
Assessment	Students estimate $\sqrt{300}$.

Assignment/HW	Estimate the square root of 145, 230, 630
Skills/Competencies	Estimation skills
Intelligences Addressed	Logical intelligence
SESSION NO. : 7	CONCEPT NAME: Long Division Method for Finding Square Roots
Learning Outcomes	Students will understand the long division method for finding square roots . Students will be able to apply the step-by-step long division process to find square roots of large numbers. Students will develop accuracy and logical reasoning while solving square root problems.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	NCERT Ganita Prakash Textbook Reference textbook Chart showing long division steps
Teacher Activity Time Assigned:	Introduction: Provocation/Recapitulation/Preassessment Teacher revises the concept of square numbers and square roots. Teacher explains that sometimes numbers are large and it becomes difficult to find the square root mentally. Teacher introduces the Long Division Method for finding square roots. Teacher demonstrates the steps with an example. Example: Find $\sqrt{576}$ Step 1: Pair the digits starting from the right $\rightarrow 5 \mid 76$ Step 2: Find the largest number whose square is less than or equal to 5 $2^2 = 4$ Step 3: Subtract and bring down the next pair of digits Step 4: Double the quotient and find the next digit Step 5: Continue the process until the remainder becomes zero. Teacher solves examples on the board: $\sqrt{576}$ $\sqrt{784}$ $\sqrt{1296}$ Methodology: Logical thinking Step-by-step demonstration Analytical reasoning
Student Activity Time Assigned: 15 min	Students observe the steps of the long division method carefully. Students practice solving square root problems using the long division method. Students solve the following problems in class: $\sqrt{256}$ $\sqrt{729}$ $\sqrt{1444}$ Students explain the steps followed in solving the problems.
Assessment	Teacher assesses students by observing their problem solving method and accuracy while applying the long division method.
Assignment/HW	Find the square roots using long division method:

	$\sqrt{2116}$ $\sqrt{3364}$ $\sqrt{4900}$
Skills/Competencies	Logical thinking Analytical thinking Problem solving
Intelligences Addressed	Interpersonal intelligence Logical-mathematical intelligence
SESSION NO. : 8	CONCEPT NAME: Application Problems on Squares
Learning Outcomes	Students apply square numbers in solving real-life problems.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook
Teacher Activity	Introduction: Provocation/Recapitulation/Preassessment Teacher explains word problems. Example: Area of square cloth = 125 cm ² Find largest square that can be cut. Methodology: Problem solving
Time Assigned	
Student Activity	Students solve textbook problems.
Time Assigned: 15 min	
Assessment	Teacher checks students’ reasoning.
Assignment/HW	Solve textbook problems
Skills/Competencies	Problem solving
Intelligences Addressed	Logical intelligence
SESSION NO. : 9	CONCEPT NAME: Introduction to Cube Numbers
Learning Outcomes	Students will understand cube numbers. Students will relate cubes to volume.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	Unit cubes
Teacher Activity	Introduction: Provocation/Recapitulation/Preassessment Teacher explains: $1^3 = 1$ $2^3 = 8$ $3^3 = 27$ $4^3 = 64$ Teacher shows cube models.
Time Assigned:	

	Methodology: Visual demonstration
Student Activity Time Assigned: 15 min	Students build cubes using unit blocks.
Assessment	Students list first 15 cubes.
Assignment/HW	Learn the cubes till 15
Skills/Competencies	Spatial reasoning
Intelligences Addressed	Visual intelligence
SESSION NO. : 10	CONCEPT NAME: Patterns in Cubes
Learning Outcomes	Students will observe patterns in cube numbers.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook
Teacher Activity Time Assigned:	Introduction: Provocation/Recapitulation/Preassessment Teacher helps students fill table of cubes. Teacher discusses patterns in unit digits.
	Methodology: Pattern observation
Student Activity Time Assigned: 15 min	Students calculate cubes from 1 to 20
Assessment	Teacher asks questions based on cube patterns.
Assignment/HW	Learn the cube patterns
Skills/Competencies	Analytical thinking
Intelligences Addressed	Logical intelligence
SESSION NO. : 11 and 12	CONCEPT NAME: Revision and Figure It Out
Learning Outcomes	Students will revise squares, cubes and square roots.
Resources Required – Physical, Digital, Books, Charts, PPT’s, Videos, Weblinks, Worksheets etc	NCERT textbook
Teacher Activity Time Assigned:	Introduction: Provocation/Recapitulation/Preassessment Teacher solves Figure It Out questions. Teacher revises key concepts of squares and cubes.
	Methodology: Discussion and revision
Student Activity	Students solve exercise questions.

Time Assigned: 15 min	
Assessment	Teacher evaluates students through problem solving.
Assignment/HW	Complete remaining textbook questions
Skills/Competencies	Logical reasoning Problem solving
Intelligences Addressed	Mathematical intelligence